RADIATION RESEARCH **142**, 16–22 (1995)

Radioprotection of Mouse Hemopoiesis by Dipyridamole and Adenosine Monophosphate in Fractionated Treatment

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Pospíšil, M., Hofer, M., Znojil, V., Vácha, J., Netíková, J. and Holá, J. Radioprotection of Mouse Hemopoiesis by Dipyridamole and Adenosine Monophosphate in Fractionated Treatment. *Radiat. Res.* 142, 16–22 (1995).

The purpose of the studies reported here was to investigate the ability of the combined administration of dipyridamole and adenosine monophosphate, drugs known to elevate extracellular adenosine, to protect mice undergoing treatment with fractionated irradiation (five doses of 2 or 3 Gy each) given at 24-h intervals. Based on observations of hemopoietic recovery (endogenous hemopoietic spleen colony formation, marrow granulocyte–macrophage colony-forming cells, peripheral blood cells) after the completion of fractionated irradiation and on survival studies, it was demonstrated that the repeated administration of the drugs 60 min before each of the radiation fractions mitigates the hemopoietic injury and enhances the survival of mice irradiated with an additional "top-up" dose. It could be deduced that the single protective actions of the drugs retain their efficacy in repeated treatment and enhance the sparing effect of dose fractionation on hemopoiesis. Interestingly, the toxic side effects of the drugs tend to decrease when they are administered repeatedly, probably due to the development of tolerance to their cardiovascular action. This reduction in toxicity offers benefit with respect to the potential use of these hemopoiesis-protecting drugs in clinical radiotherapy.

INTRODUCTION

Recently, we have described a radioprotective mechanism based on the signaling by extracellular adenosine (*1–5*). It was shown that combined treatment with dipyridamole, which inhibits the uptake of adenosine, and a soluble adenosine prodrug, adenosine monophosphate (AMP), caused an increase of extracellular adenosine. This treatment protected mice subjected to single radiation exposures. Based on survival studies, our pilot experiments have shown that these protective mechanisms can also be effective when they are induced repeatedly in the course of fractionated irradiation using nine 3-Gy fractions delivered two times weekly (*3*).

The purpose of the experiments reported here was to investigate in more detail the ability of the combined administration of dipyridamole and AMP to protect mice in the course of fractionated treatment. The fractionation regimen consisted of five radiation doses given at 24-h intervals. Such an experimental model is analogous to the standard fractionation schedule used in clinical radiotherapy. The drugs were administered 60 min before each radiation fraction, i.e. in a treatment scheme proposed to stimulate the proliferation of hemopoietic cells and to avoid hypoxia (*3*). Hemopoietic recovery and survival of animals were used as indices of radiation damage. The experimental protocol also included studies evaluating the risk of toxicity of the drugs when administered repeatedly.

MATERIALS AND METHODS

Mice

Male (CBA × C57BL/10)F_1 mice 3 months old and weighing an average of 25 g were used. A standardized pelleted diet and tap water treated with HCl (pH 2–3) were given *ad libitum*. The mice were kept under controlled lighting conditions (light:dark 12:12) and the temperature was maintained at 22 ± 1°C.

Irradiation

Mice were given total-body doses from a ^{60}Co γ-ray source at a dose rate of 0.3 Gy/min. During irradiation the mice were placed in ventilated Plexiglas containers. Single and fractionated doses were used. In the fractionated regimen mice were given equal daily fractions of 2 or 3 Gy. Five fractions of 3 Gy were not lethal; no deaths of mice were recorded within 30 days after the completion of irradiation. To investigate radiation damage in terms of lethality, 24 h after the last fraction the mice were exposed to an additional "top-up" dose of 3.5 Gy.

Protective Treatment with Drugs

A protective combination of dipyridamole and adenosine monophosphate (AMP), at doses shown to be effective in our earlier experiments with single radiation exposures (*3*), was used. Because the differences in the initial body weight of the animals used in the experiment were small (±2 g), the drug doses were given on a per mouse basis. Dipyridamole (Sigma) was dissolved in 0.4% tartaric acid and injected subcutaneously (sc) at a dose of 2 mg per mouse in a volume of 0.4 ml. Adenosine 5′-monophosphate sodium salt from yeast (Sigma) was dissolved in distilled water and injected intraperitoneally (ip) at a dose of 5 mg free acid per mouse in a volume of 0.2 ml. Dipyridamole was administered 20 min before AMP; AMP was always given 60 min before irradiation. Tartaric acid (0.4%) and saline were used for control sc and ip injections, respectively. As found earlier (*3*), a 50% acute, 2-day lethality can be induced

0033-7587/95 $5.00

in normal mice by the combination of 2 mg of dipyridamole and 75 mg of AMP. Thus, with respect to AMP, the single protective dose used in our experiments represented $^1/_{15}$ of the $LD_{50/2\ days}$.

Measurements of Core Temperature

Core temperature was measured in conscious mice at a room temperature of 22°C with a thermistor thermometer. The optimal duration and depth of probe insertion was 1 min at 1.7 cm into the rectum.

Hematological Methods

Endogenous spleen colony-forming units (CFU-S) were determined on day 10 after single or fractionated irradiation. Animals were euthanized by cervical dislocation, their spleens were removed and fixed in Bouin's solution, and grossly visible nodules larger than 0.4 mm in diameter were counted (*6*). In some experiments, the spleens were embedded in paraffin. A single midline longitudinal section was considered to be a representative sample of each spleen (*7*). Using a light microscope, colonies were counted in four size categories: I, <0.25 mm (more than 10 cells were considered to be a colony); II, 0.25–0.5 mm; III, 0.5–1.0 mm; and IV, >1.0 mm. Average colony volumes (mm^3) were estimated for each spleen. The calculations used were a modification of those of Hellman and Grate (*8*), when numbers of colonies in individual size categories were taken into account.

Hemopoietic progenitor cells committed to development of granulocytes and/or macrophages (GM-CFC) were assayed in the femoral marrow by a semi-solid plasma-clot technique (*9*). Bone marrow cell suspensions were plated in quadruplicate using 10% murine lung conditioned medium as a source of colony-stimulating factor. Colonies (>50 cells) were counted after 7 days of incubation in a 37°C humidified environment containing 5% CO_2.

In addition, spleen weight, bone marrow cellularity and the numbers of blood leukocytes and erythrocytes were determined. For differential counting of leukocytes, blood smears were stained by the May-Grünwald and the Giemsa-Romanowski method. The nucleated cells in the femur and peripheral blood cells were enumerated using a Coulter Counter. Blood was withdrawn from a fine incision in the tail vein.

Survival Assay

In lethally irradiated mice, deaths were recorded daily up to day 30 after exposure. The percentage of mice surviving 30 days was used in the analysis of the effect of the treatments.

Statistics

Results of replicate experiments were pooled and represent the mean ± standard error (SEM) of pooled data. Simultaneous investigations of two differently treated groups and one control group were performed in the case of multiple comparisons. Before the results of repeated experiments were pooled, the homogeneity of the results was verified. The unpaired Student's *t* test (where appropriate, Dunnett's tables for multiple comparisons with a control were applied), analysis of variance and nonparametric Mann-Whitney rank sum test were used to determine statistical differences in all but survival data. The statistical treatment of the data for the CFU-S took into account the error of colony overlap that is inherent in the method (*10*). Dose–survival curves for spleen hemopoietic colonies were fitted by the weighted least-squares method, and the analysis of variance was used to evaluate the significance of the protective effects. Survival data were analyzed using the χ^2 test with the Yates correction. The level of significance was set at $P < 0.05$.

RESULTS

Toxicity of the Combination of Dipyridamole and AMP when Administered Repeatedly

Acute, 2-day lethality was investigated in mice irradiated with five daily 3-Gy doses, treated repeatedly with the vehicle or protective doses of drugs, and injected 6 h after the completion of the radiation regimen with an additional toxic dose of drugs, i.e. with 2 mg of dipyridamole combined with 75 mg of AMP. While a 48% lethality (23 animals in group) occurred in mice treated only with the high dose of drugs, no deaths were recorded in mice (24 animals in group) pretreated with five protective doses of drugs and injected with the additional toxic dose. The difference was significant at the 0.001 level (χ^2 test). Thus there was no evidence of cumulative toxicity; on the contrary, the results implied development of tolerance to the toxicity of the drugs. The dose of 75 mg of AMP in combination with 2 mg of dipyridamole was found in our previous experiments to represent the $LD_{50/2\ days}$ in nonirradiated normal mice (*3*). The comparison of this value with the results obtained in vehicle-treated irradiated mice (48% lethality within 2 days) suggested that irradiation *per se* did not modify the acute toxicity of the combination of drugs.

As shown earlier (*3*), hypothermia in response to dipyridamole + AMP is a typical manifestation of the action of the drugs and can be attributed to the development of systemic hypotension and hypoxia, i.e. effects closely connected with the toxicity of the drugs. The question was asked whether the decrease in core temperature could be modified by the repeated treatment with the drugs. Mice irradiated with four 3-Gy doses were either protected before each fraction with dipyridamole + AMP or treated only with the vehicle. One day after the last irradiation, mice of both groups were injected with the protective dose of the drugs (without irradiation) and the core temperature was measured. The results of this experiment are given in Fig. 1. In mice receiving the drugs only once, there was a pronounced drop in temperature with a nadir at 1 h and recovery up to 3 h after treatment. The hypothermia in animals subjected to the repeated administration of the drugs was significantly less pronounced and the temperature returned to normal by 2 h after treatment. Thus the data indicated that repeated administration of the drugs induced an enhanced tolerance to hypothermia.

The Effects of the Repeated Dipyridamole + AMP Pretreatment on the Spleen Response after Single-Dose Irradiation

With respect to the findings suggesting the development of tolerance to the toxic effects of the drugs, experiments were performed to investigate the modification of the protective effect of dipyridamole + AMP by their repeated administration. Splenic response on day 10 after a single dose of 6.5 Gy was used as a criterion of the protective effect. Three groups of mice were compared: control vehicle-treated mice, animals treated with dipyridamole + AMP 60 min prior to irradiation, and animals treated with the drugs 60 min and 1, 2, 3 and 4 days before irradiation. As shown in Table I, single administration of the drugs prior to irradiation enhanced the

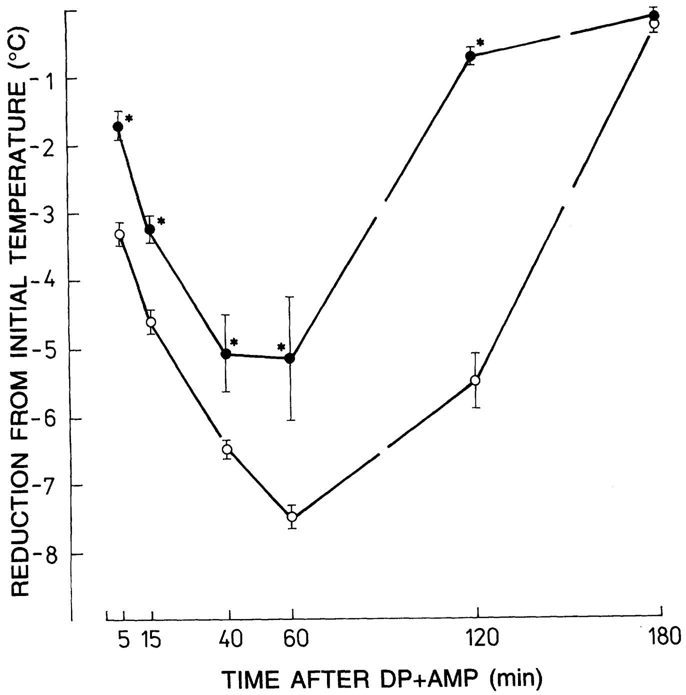

FIG. 1. Changes in rectal temperature expressed as differences from the initial value, measured up to 180 min after treatment with protective doses of dipyridamole (DP) + AMP. Hypothermia in response to the drugs was measured 24 h after the completion of fractionated irradiation (four daily 3-Gy doses) in mice that were either protected before each radiation dose with dipyridamole + AMP (●) or treated only with the vehicle (○). Asterisks denote statistical significance ($P < 0.01$) when compared to the group injected with the drugs only once (Mann-Whitney rank sum test). Data are expressed as means ± SEM for six measurements.

colony formation and the spleen weight significantly compared to untreated irradiated controls. Priming of the mice with four doses of the drugs elevated the counts of spleen colonies and spleen weight significantly. Such a stimulatory effect of the priming treatment was also observed when evaluating the microscopically counted spleen colonies and their average volumes, suggesting an enhanced proliferation and/or differentiation of hemopoietic cells. Thus, in terms of the hemopoiesis-stimulating effect, repeated administration of dipyridamole + AMP did not induce the development of tolerance; on the contrary, accumulation of the beneficial hemopoiesis-stimulating action operating independently on the radiation damage was observed.

Radioprotective Efficacy of Dipyridamole + AMP Administered Repeatedly: Fractionated Irradiation

Hemopoietic damage. The effects of the repeated protective administration of dipyridamole + AMP were investigated in mice given five 3-Gy doses of radiation. Various hematological indices were examined on day 10 after the completion of fractionated treatment and compared with those obtained in control vehicle-treated mice and mice protected with the drugs only once before the last radiation fraction. Data shown in Tables II and III indicate that the repeated administration of the drugs reduced significantly the severity of damage to hemopoietic tissue. This effect was evident in the stem cell compartments (CFU-S, GM-CFC), in both the murine hemopoietic organs (bone marrow, spleen) and the counts of blood cells. A single administration of the protective drugs did not induce such effects. The beneficial action of the single protective treatment was manifested only by a slight but significant elevation of the GM-CFC counts in the bone marrow.

Figure 2 illustrates the postirradiation response of the peripheral blood cells in mice given dipyridamole + AMP before each of the five daily 2-Gy doses. A significant acceleration of the recovery of the granulocyte, lymphocyte and erythrocyte counts was found in protected mice on day 10 after the completion of fractionated irradiation. Blood granulocytes in the protected mice exceeded the normal

TABLE I
Spleen Response on Day 10 after 6.5 Gy Single Irradiation in Groups of Control Mice and Mice Pretreated with Dipyridamole (DP) + AMP Given either Once or Five Times

	Spleen weight (mg)	Grossly visible colonies	Histologically evaluated colonies	Average colony volume of histologically evaluated colonies (mm^3)
Irradiated control	23.9 ± 0.8 (11)	2.2 ± 0.3 (11)	not done	not done
DP + AMP 1× (60 min before irradiation)	33.6 ± 0.9 (12)^a	6.7 ± 0.8 (12)^a	5.8 ± 1.0 (6)	0.05 ± 0.01 (6)
DP + AMP 5× (60 min, and on days 1, 2, 3 and 4 before irradiation)	48.2 ± 2.6 (13)^{a,b}	10.4 ± 0.8 (13)^{a,b}	9.5 ± 1.0 (6)^b	0.28 ± 0.02 (6)^b

Notes. Data represent means ± SEM of values obtained from two experiments. Numbers in parentheses are numbers of animals. Mann-Whitney rank sum test was used to determine statistical differences; for the sake of simplicity an 0.05 level of significance was used for all comparisons.
$^a P < 0.05$ compared to irradiated control.
$^b P < 0.05$ compared to group DP + AMP 1×.

TABLE II
Hematological Indices on Day 10 after Completion of Fractionated Irradiation (Five Daily 3-Gy Doses) in Vehicle-Treated Control Mice and Mice Protected with Dipyridamole (DP) + AMP Given either Once or Five Times, and in Nonirradiated Controls

	Nucleated cells per femur $\times 10^7$	GM-CFC per femur	Leukocytes per μl of blood	Erythrocytes per μl of blood $\times 10^6$
Irradiated control	0.74 ± 0.03 (34)	285 ± 29 (17)	537 ± 33 (34)	5.51 ± 0.13 (34)
DP + AMP 1× (before last radiation fraction)	0.83 ± 0.07 (14)	823 ± 72 (14)[a]	547 ± 70 (14)	5.50 ± 0.16 (14)
DP + AMP 5× (before each radiation fraction)	1.35 ± 0.04 (40)[a,b]	8,603 ± 513 (14)[a,b]	1,446 ± 102 (40)[a,b]	6.61 ± 0.10 (40)[a,b]
Nonirradiated control	2.55 ± 0.14 (15)	20,393 ± 1,152 (15)	6,440 ± 480 (15)	10.20 ± 0.20 (15)

Notes. Data represent means ± SEM of values obtained from two to four experiments. Numbers in parentheses are numbers of animals. Student's *t* test was used to determine statistical differences.
[a] $P < 0.01$ compared to irradiated control.
[b] $P < 0.01$ compared to group DP + AMP 1×.

values by 69%, while those observed in control mice attained only 58% of the norm. The differences between control and protected mice disappeared by day 17 postirradiation, when counts of blood cells in the control group returned to near-normal levels (data not shown).

The response of grossly visible endogenous spleen colonies was evaluated in vehicle-treated and repeatedly protected mice subjected to three, four and five daily 3-Gy fractions. These effects have been compared with our earlier results describing the modification of single dose–survival curves for CFU-S by a single administration of dipyridamole + AMP (*3*). The results are depicted in Fig. 3. The dose response to fractionated irradiation of 3 Gy daily in control mice clearly differed from the single dose–survival curve for controls; the shape of the curve, i.e. its more shallow slope, suggested the known sparing effect of fractionated irradiation on CFU-S (*11*). The endogenous spleen colony formation in repeatedly protected mice was enhanced further compared to controls. Data have been subjected to analysis of variance which involved both the single and fractionated irradiation regimens and was based on the verified assumption that all the experimental subsets had the same dispersions. A general linear regression model was used encompassing eight terms:

$$y = a_o + a_f f + a_x x + a_p p + a_{fx} fx + a_{fp} fp + a_{xp} xp + a_{fxp} fxp,$$

where coefficients a_i are values to be determined, y denotes the logarithm of colony number, f represents the radiation regimen (single dose vs fractionation; 0 or 1), p characterizes protective treatment (0 or 1), and x is the radiation dose. The following three terms were found not to contribute to the overall variance and were excluded: $a_p p$, $a_{fp} fp$ and $a_{fxp} fxp$. In this way, the following common equation for all experimental data was obtained:

$$y = 8.98679 - 5.21217f - x(1.21078 - 0.96306f - 0.12106p).$$

The four regression lines demonstrated in Fig. 3 correspond to the equation for the given combination of f and p

TABLE III
Spleen Response on Day 10 after Completion of Fractionated Irradiation (Five Daily 3-Gy Doses) in Vehicle-Treated Control Mice and Mice Protected with Dipyridamole (DP) + AMP Given either Once or Five Times

	Spleen weight[c] (mg)	Grossly visible colonies	Histologically evaluated colonies
Irradiated control	21.2 ± 0.6 (34)	1.0 ± 0.2 (23)	3.3 ± 0.6 (8)
DP + AMP 1× (before last radiation fraction)	22.8 ± 0.6 (14)	1.1 ± 0.2 (14)	1.4 ± 0.3 (14)
DP + AMP 5× (before each radiation fraction)	30.5 ± 1.2 (40)[a,b]	6.5 ± 0.7 (26)[a,b]	24.7 ± 2.9 (8)[a,b]

Notes. Data represent means ± SEM of values obtained from two to four experiments. Numbers in parentheses are numbers of animals. Statistical differences were determined using Student's *t* test (spleen weight) and Mann-Whitney rank sum test (spleen colonies).
[a] $P < 0.01$ compared to irradiated control.
[b] $P < 0.01$ compared to group DP + AMP 1×.
[c] Spleen weight in nonirradiated control mice: 72.0 ± 3.2 mg (*15*).

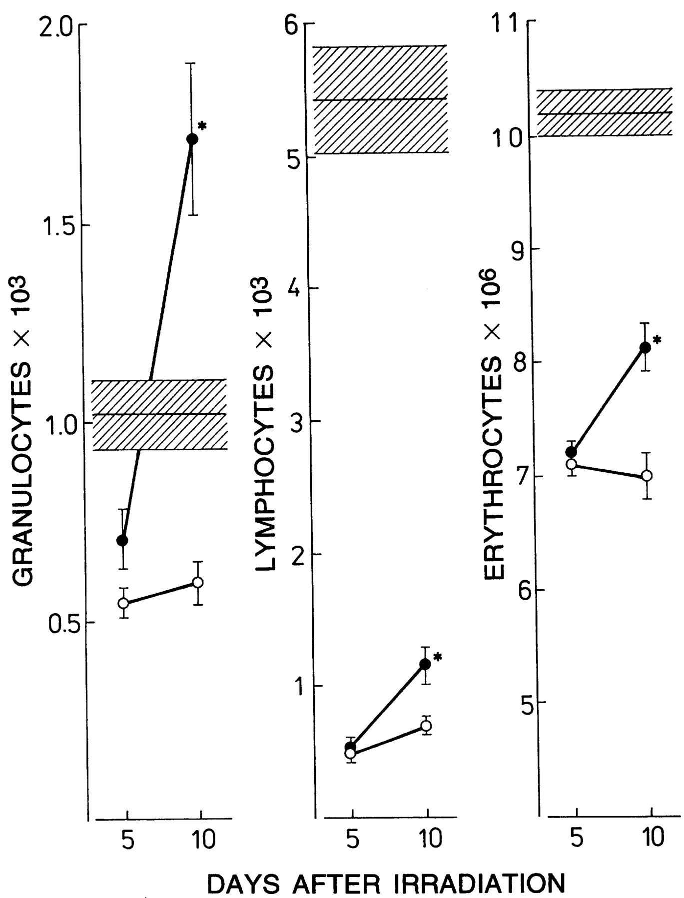

FIG. 2. Peripheral blood cell counts (per μl) determined on days 5 and 10 after the completion of fractionated irradiation (five daily 2-Gy doses) in mice treated before each of the radiation fractions with dipyridamole + AMP (●) and in vehicle-treated control mice (○). The hatched areas represent the values obtained in nonirradiated mice. Asterisks denote statistical significance ($P < 0.01$) when compared to vehicle-treated mice (Student's t test). Data are expressed as means ± SEM for 12–14 measurements.

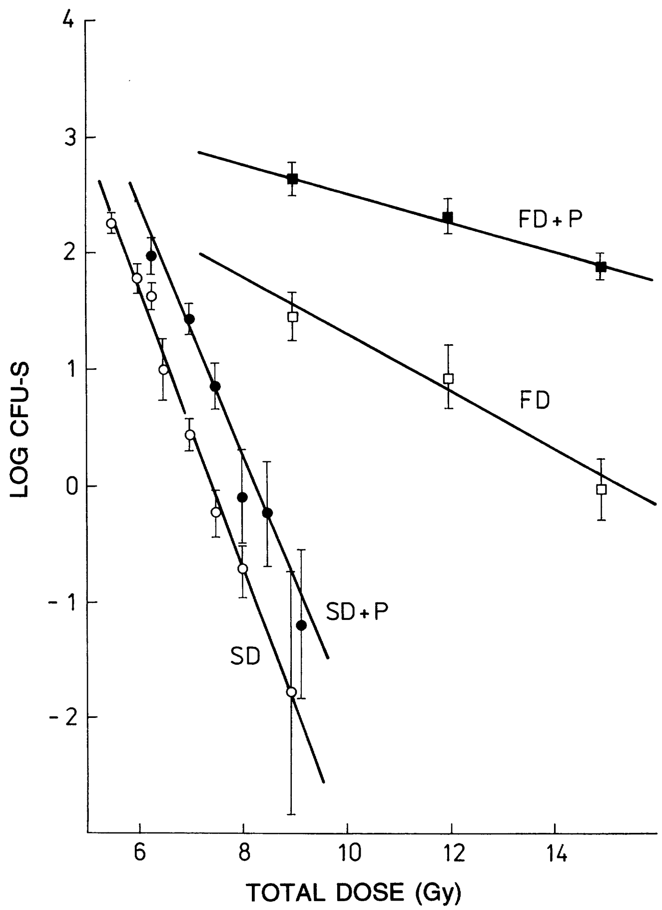

FIG. 3. Dose–survival curves for endogenous CFU-S in mice exposed to single radiation doses (SD) or 3-Gy daily dose fractions (FD). Mice were either protected with the combination of dipyridamole + AMP (P) given before each irradiation or treated only with the vehicle. Results are shown as means ± SEM. Ten to 40 mice per point were used. For statistical analysis see Results.

values. The residual variance is approximately equal to the value expected on the basis of colony-counting precision (*10*), which suggests the absence of any other uncontrolled effect and of nonlinearity. Only highly statistically significant terms are represented in the equation, with the lowest F value being 182.548 ($P < 10^{-5}$). This statistical analysis provided the following information: (1) The effect of the protective treatment is highly significant. (2) The protective effect appears only in connection with the dose in the equation given above; i.e., it does not depend on the irradiation regimen. It is apparent from Fig. 3 that the protective effect increases with increasing dose, as the curves for the protected and nonprotected animals diverge (in both irradiation regimens). (3) The equal-effect dose ratios are higher for the fractionated irradiation compared to the single-dose irradiation. It follows that the effect of the protective treatment is more pronounced in the fractionated regimen ($F = 10.959$, $P < 0.005$).

Postirradiation survival. Vehicle-treated or repeatedly protected mice were irradiated with five daily 3-Gy doses and exposed to an additional "top-up" dose of 3.5 Gy 24 h after the last fraction (without protection). The data shown in Fig. 4 demonstrate a significant protective effect of the combination of dipyridamole + AMP administered repeatedly in terms of postirradiation survival.

DISCUSSION

The present study demonstrates that repeated administration of the combination of dipyridamole and AMP in a fractionated irradiation regimen enhances postirradiation hemopoietic recovery in sublethally irradiated mice and promotes survival of mice subject to an additional lethality-

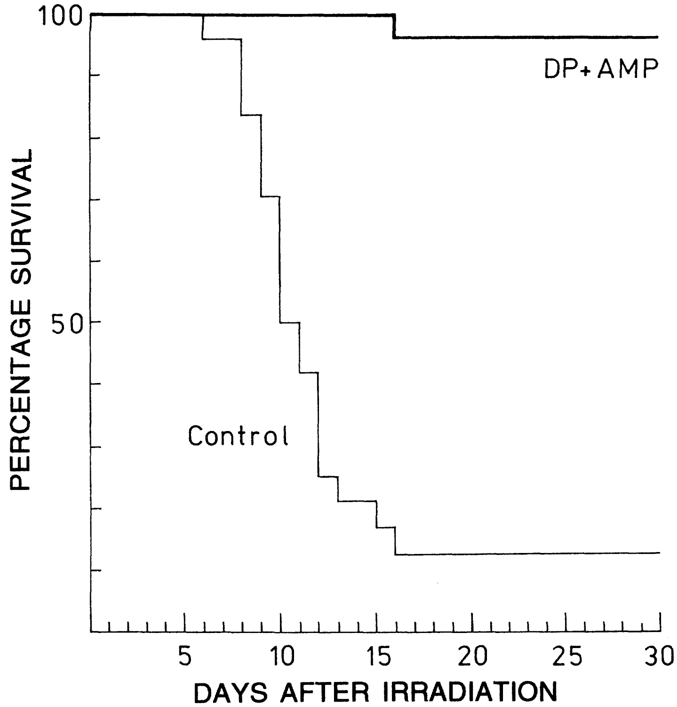

FIG. 4. Survival of mice irradiated with five daily 3-Gy fractions and exposed 24 h after the last fraction to a "top-up" dose of 3.5 Gy. Mice were either protected with dipyridamole (DP) + AMP before each of the five radiation fractions or treated only with the vehicle (control). Data were obtained from four replicate experiments. The protected group comprised 27 mice, the control group 24 mice. The difference in 30-day survival is significant at the 0.001 level (χ^2 test).

inducing radiation dose. The protective hemopoiesis-stimulating effects were manifested at the level of pluripotent stem cells (CFU-S), committed stem cells (GM-CFC) and mature white and red blood cells. The enhanced survival observed in drug-treated mice was probably due to accelerated hemopoietic regeneration.

The radioprotective mechanisms of the drug action are proposed to act via elevation of the extracellular adenosine levels and activation of the adenosine cell receptors (*1–3*). As has been shown in our earlier paper describing the protective efficacy of dipyridamole + AMP after a single radiation exposure (*3*), the action of the drugs can mediate at least two physiological effects that may be viewed as radioprotective ones. Bradycardia and vasodilation, induced by the elevation of extracellular adenosine, can lead to short-lasting hypotension and hypoxia in radiosensitive tissues and be radioprotective via the oxygen effect. Such a protective mechanism occurred if the animals were irradiated 15 min after administration of the drugs and was manifested by a decrease in the radiosensitivity of hemopoietic stem cells (an increase in the D_0). Direct measurements of oxygen tension in tissues have shown that 60 min after administration of the drugs tissue hypoxia fades away; the D_0 values of hemopoietic stem cells do not change, but the drug action at 60 min preirradiation enhances proliferation in hemopoietic cell compartments and is also radioprotective. Hypothermia in mice, as used in the present experiments evaluating the toxic action of the drugs, can be judged as a prolonged compensatory reaction of the organism to hypoxia and hypometabolism (*12*) and does not need to be correlated with time in the development of hypoxia. Earlier radiobiological experience suggests that hypothermia *per se* cannot be considered to be radioprotective (*13*, *14*). Additional support of the concept of the role of the two independent radioprotective mechanisms of the action of the drugs was obtained in our experiments (*15*) showing that the joint use of an optimal dose of noradrenaline with the combination of dipyridamole + AMP eliminated the cardiovascular and hypoxic effects of the treatment but preserved the ability of the drugs to enhance postirradiation hemopoietic recovery. Taking these suggestions into account, the interpretation of our results is based on the assumption that, under the experimental conditions used, extracellular adenosine acts as a mediator in the stimulation of proliferation of hemopoietic cells. The administration of the drugs 60 min prior to irradiation was also substantiated by our earlier results demonstrating a more effective protection in terms of postirradiation survival in mice exposed to fractionated irradiation (9 × 3 Gy twice weekly) and receiving the drugs 60 min compared to 15 min before radiation treatment (*3*).

Even though the signaling pathways of the proposed enhancement of cell proliferation by the drugs remain to be analyzed, the elevation of extracellular adenosine seems to act as a physiological stimulant of proliferation of hemopoietic cells. This was suggested by our earlier findings demonstrating hemopoiesis-enhancing effects of the drugs in normal nonirradiated mice (*2*), as well as by the results presented here showing enhanced formation of endogenous spleen colonies in mice irradiated with a single exposure and primed before irradiation with four daily doses of the drugs. The enhancement of cell proliferation in cell compartments of hemopoiesis might be a reasonable explanation of the protective effects observed. This protective mechanism probably acts additively with the compensating proliferation pressures induced by the radiation damage to cells, which operate under conditions of fractionated irradiation delivered at 24-h intervals (*11*). However, the role of other mechanisms cannot be excluded. We have shown recently that postirradiation administration of dipyridamole + AMP to mice irradiated with a low dose of 1 Gy reduces the early postirradiation damage to sensitive thymocytes and enhances the rejoining processes of DNA strand breaks (*5*). The role of these mechanisms of the drug action in the response of the whole hemopoietic system to irradiation remains to be elucidated.

Our results suggest that the single protective actions of the drugs on hemopoiesis retain their efficacy in repeated treatments, and that the accumulation of the single protective effects could be assumed for the explanation of the manifestations of protection in the fractionated regimen. Such a conclusion is based on the results demonstrating the priming action of the repeated drug administration on the postirradiation splenic response when using single radiation exposures, and the absence of the protective effect in animals receiving the drugs only once when compared to the efficacy of the repeated treatment. This conclusion was also corroborated by the comparison of the shapes of the survival curves for CFU-S in control and protected mice, both irradiated with three, four and five fractions. If the single protective actions had gradually lost their efficacy, this could have been observed as an increase in the slope of the survival curve for CFU-S.

An interesting finding of our studies was the development of tolerance to the toxic effects of the drugs. The existence of such a physiological adaptive mechanism offers benefit with respect to the potential clinical use of the drugs in radioprotection. Since the toxic effects of dipyridamole + AMP can be attributed to the vasodilatory and thus hypotensive action of extracellular adenosine, mediated via cell surface A_2 receptors on the vascular smooth muscle (*16*), development of tolerance to these effects could be induced by a down-regulation of the receptors and/or stimulation of compensatory systems, such as the renin angiotensin system and sympathetic activation (*17*).

To our knowledge, this is the first study to demonstrate the possibility of using protective agents acting via stimulation of hemopoiesis in standard 24-h fractionated irradiation regimens. In this context, only attempts to use the granulocyte colony-stimulating factor in the treatment of leukopenia during fractionated radiotherapy can be mentioned (*18*). Thus our results might contribute to the open problem of radioprotection of normal tissues in radiotherapy. However, before considering the applied aspects of these findings, experiments evaluating the action of the drugs on the responses of tumors to irradiation are needed.

ACKNOWLEDGMENTS

This work was supported by Grant No 306/93/0051 obtained from the Grant Agency of the Czech Republic. The research was conducted according to the principles enunciated in the "Guide for the Care and Use of Laboratory Animals" issued by the Czech Society for Laboratory Animal Sciences.

Received: April 4, 1994; accepted: December 19, 1994

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